DESIGN PATENT DOCUMENTATION

PLASMA ARC METAL RECOVERY SYSTEM

Ultra-High-Temperature Circular Material Recovery System for Electronic Waste

Title	Plasma Arc Metal Recovery System — High-Temperature Circular Material Recovery System for Electronic Waste
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Statement of Originality

The originality of the proposed design lies in the transformation of a conventional linear process poster into a proprietary sustainability-driven visual architecture representing an advanced plasma-based metal recovery operating system. Unlike traditional educational infographics that rely on dense, multi-stage technical blocks, the proposed design introduces a simplified five-stage recovery pipeline, a central ionized plasma reactor visualization, a recovered-metal identification system, an industrial process-analytics dashboard, and a clean white-and-green sustainability interface language.

The novelty is derived from the unique combination of:

- Simplified five-stage plasma recovery pipeline architecture
- Central Plasma Arc Reactor Core
- Ultra-high-temperature ionized plasma visualization (5,000–15,000 °C)
- Recovered-metal identification module
- Sustainability-oriented white-and-green interface design
- Industrial process-highlight dashboard
- Circular reuse and recycling iconography
- Compact, low-maintenance operational framing

The design creates a distinctive visual identity that resembles a clean, cost-efficient industrial recovery system rather than a conventional dense academic infographic.

Figure 1 — Perspective View

Figure 1 illustrates the complete perspective view of the proposed Plasma Arc Metal Recovery System.

The design consists of a large centralized reactor pathway positioned at the core of the visual architecture. Surrounding the reactor is a left-to-right process framework that represents the complete lifecycle of valuable metal recovery from PCB-based electronic waste. The process flow includes:

1. Electronic Waste Input
2. Plasma Arc Smelting
3. Gas Cleaning & Filtering
4. Metal Collection
5. Reuse & Recycling

The design communicates a continuous, cost-efficient resource recovery ecosystem rather than a complex, multi-stage extraction process.

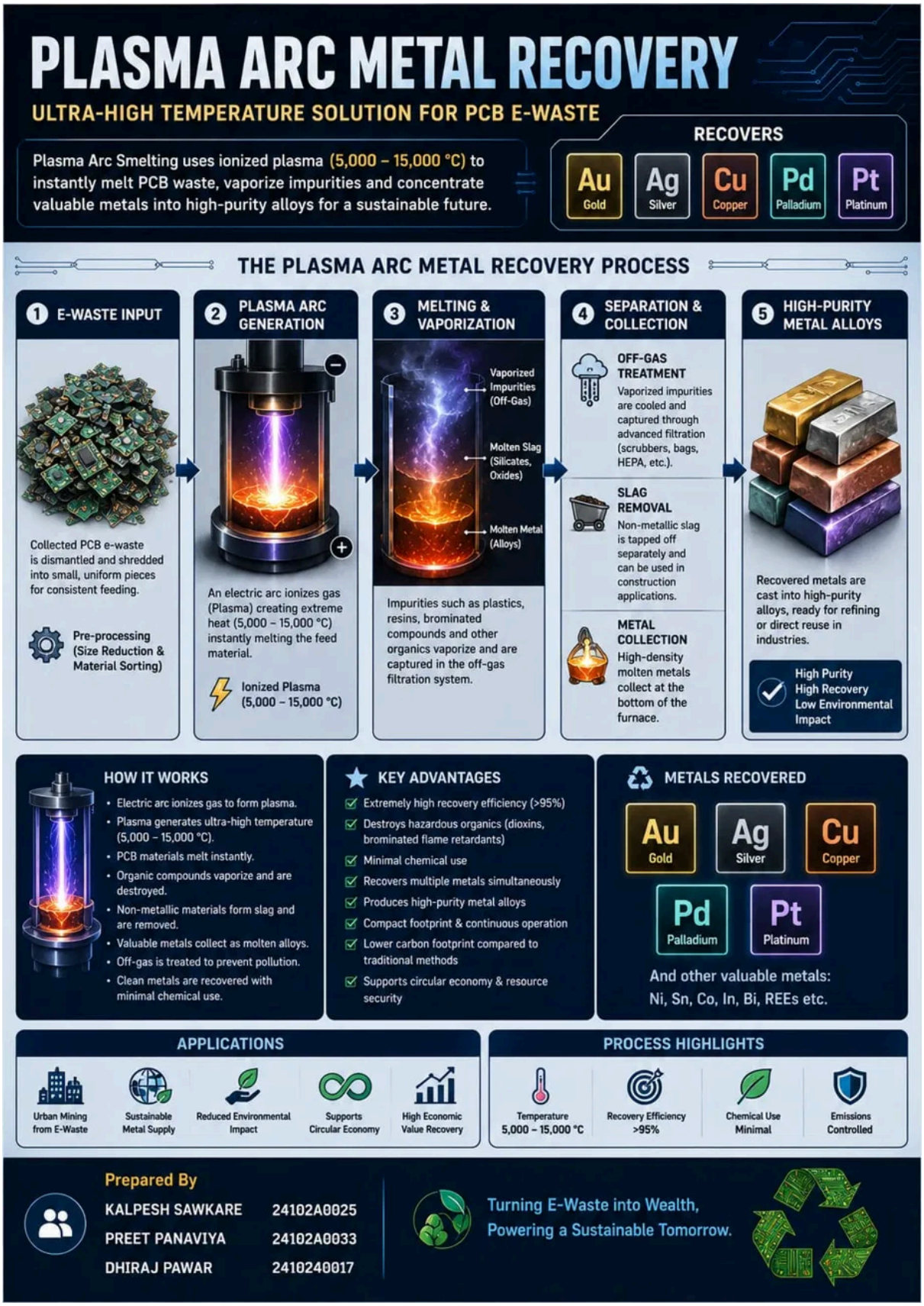


FIG. 1 — Perspective view of the Plasma Arc Metal Recovery System.

Figure 2 — Central Reactor View

Figure 2 illustrates the central plasma reactor subsystem, which serves as the visual anchor and operational core of the entire recovery architecture. Three key operations surround the reactor:

1. Plasma Arc Generation

Responsible for ionizing gas to create an ultra-high-temperature arc.

- Operating range: 5,000 – 15,000 °C

2. Melting & Vaporization

Responsible for instantly melting the PCB feed material and vaporizing non-metallic impurities.

- Molten metal (alloys)
- Vaporized impurities (off-gas)
- Molten slag (silicates, oxides)

3. Separation & Collection

Responsible for off-gas treatment, slag removal, and collection of high-density molten metals.

- Gold (Au)
- Silver (Ag)
- Copper (Cu)
- Palladium (Pd)
- Platinum (Pt)

All subsystems are connected through a clean, compact process pathway represented as simplified directional arrows.



FIG. 2 — Central Plasma Reactor and surrounding recovery subsystems.

Figure 3 — Recovered Metal Network

Figure 3 illustrates the recovered-metal identification architecture. A dedicated "Valuable Metals Recovered" panel surrounds the central reactor pathway. Recovered metals are displayed as clearly labelled identification tiles:

Recovered Metals

- Au — Gold
- Ag — Silver
- Cu — Copper
- Pd — Palladium
- Pt — Platinum

The tile configuration creates the primary visual identity of the design and serves as a clear, at-a-glance representation of continuous metal recovery.

Figure 4 — Process Flow Network

The design incorporates a simplified left-to-right routing framework that visualizes the recovery process as connected stages joined by directional arrows.

Stage Connectors

- E-Waste Input → Plasma Arc Smelting
- Plasma Arc Smelting → Gas Cleaning & Filtering
- Gas Cleaning & Filtering → Metal Collection
- Metal Collection → Reuse & Recycling

These pathways connect the source material, the plasma reactor, the filtration system, the collection stage, and the final reuse destination.

Figure 5 — Process Analytics Dashboard

The design integrates a compact process-highlight panel displaying key operational intelligence. Dashboard modules include:

- Temperature (5,000 – 15,000 °C)
- Recovery Efficiency (>95%)
- Chemical Use (Minimal)
- Emissions (Controlled)
- Operation (Compact, safe, automated)
- Economic Value (Turns e-waste into wealth)

The monitoring system utilizes simple iconography, clean data tiles, and a sustainability-focused colour-coded panel layout.

Figure 6 — Applications & Sustainability Ecosystem

The lower section of the design replaces a conventional process timeline with an applications and sustainability panel. The ecosystem includes:

- Urban Mining from E-Waste
- Sustainable Metal Supply
- Reduced Environmental Impact
- Circular Economy Support
- High Economic Value Recovery

A closing "Greener Tomorrow" panel visualizes recovered value being reinvested into a sustainable future, creating a complete end-to-end recovery narrative.

Distinctive Visual Features

The proposed design introduces the following proprietary visual innovations:

1. Central Plasma Arc Reactor Visualization
2. Simplified Five-Stage Recovery Pipeline
3. Recovered-Metal Identification Tiles
4. Sustainability-Oriented White-and-Green Interface
5. Compact Process-Highlight Dashboard
6. Clean Directional Flow Iconography
7. Applications & Circular Economy Panel
8. Low-Complexity Industrial Command Layout
9. Rounded Card-Based Information Architecture
10. Greener-Tomorrow Sustainability Closing Panel

Visual Style Specification

Background

- Clean White Base
- Soft Green Gradient Accents
- Minimal Card Shadowing

Accent Colors

- Forest Green
- Industrial Gold
- Slate Grey
- Copper Orange

Visual Effects

- Soft Drop Shadows
- Rounded Card Panels
- Simple Directional Arrows
- Flat Iconography
- Light Ambient Illumination
- Minimalist Industrial Depth

Final Generation Prompt

Create a portrait-oriented, patent-grade, conference-quality scientific infographic titled "PLASMA ARC METAL RECOVERY SYSTEM — Ultra-High-Temperature Circular Material Recovery System". Use a clean white background with a simplified left-to-right five-stage process flow: E-Waste Input, Plasma Arc Smelting, Gas Cleaning & Filtering, Metal Collection, and Reuse & Recycling. Integrate a valuable-metals identification panel (Au, Ag, Cu, Pd, Pt), a compact process-highlight dashboard (temperature, recovery efficiency, chemical use, emissions, operation, economic value), a "Why Plasma Arc" benefits panel, and an applications and sustainability closing section. Use white and green sustainability tones with gold, copper, and slate accent colours. Add soft card shadows, rounded panels, and flat, easy-to-read iconography. The final design must resemble a clean, cost-efficient, sustainability-oriented industrial recovery system rather than a dense technical schematic, while preserving all scientific content and recovery stages from the original infographic.